

Sec 4.1 – Using First and Second Derivatives

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Let's review what f' and f'' tell us about the graph of f

Suppose f is a differentiable function.

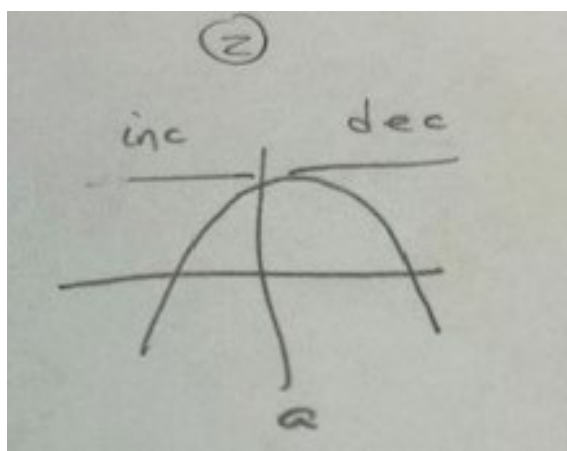
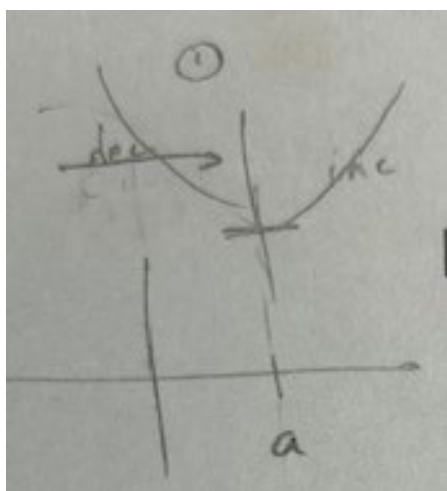
What does each of the following tell us about the graph of f ?

- $f'(x) > 0$ f is increasing
- $f'(x) < 0$ f is decreasing
- $f'(x) = 0$ f has a horizontal tangent line
- $f''(x) > 0$ f is concave up
- $f''(x) < 0$ f is concave down

Partner up

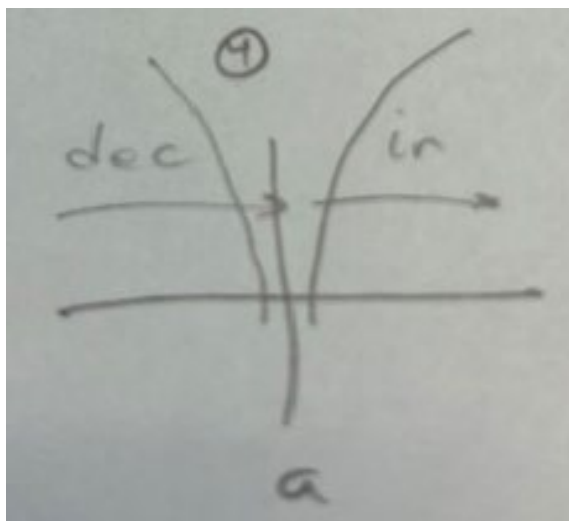
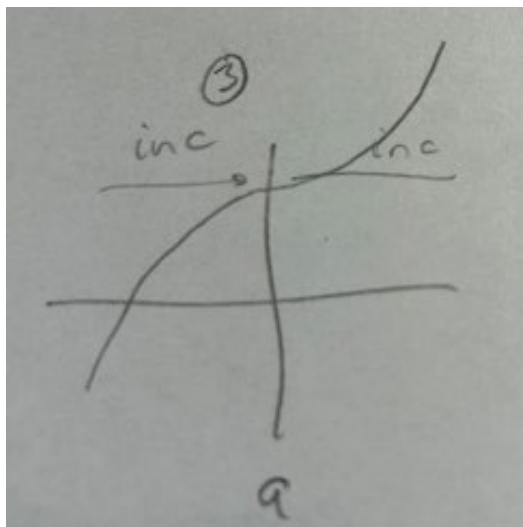
1. For $x < a$, $f'(x) < 0$;
 $f'(a) = 0$;
for $x > a$, $f'(x) > 0$

2. For $x < a$, $g'(x) > 0$;
 $g'(a) = 0$;
for $x > a$, $g'(x) < 0$



3. For $x < a$, $h'(x) > 0$;
 $h'(a) = 0$;
 for $x > a$, $h'(x) > 0$

4. For $x < a$, $k'(x) < 0$;
 $k'(a)$ DNE; for $x > a$,
 $k'(x) > 0$



In each of our sketches on the last slide, a is a *critical point* of the function:

Definition

For any function f , a point p in the domain of f where

$$f'(p) = 0 \quad \text{or} \quad f'(p) \text{ is undefined}$$

is called a **critical point** of the function. In addition, the point $(p, f(p))$ on the graph of f is also called a **critical point**. A **critical value** of f is the value, $f(p)$, at a critical point, p .

Definition (Local Extrema)

Suppose p is a point in the domain of f :

- f has a **local minimum** at p if $f(p)$ is less than or equal to the values of f for points near p .
- f has a **local maximum** at p if $f(p)$ is greater than or equal to the values of f for points near p .

Some sources refer to local extrema as *relative* extrema.

- Look back at your sketches and label all local maxima and minima.
- What connection(s) do you see between critical points and local maxima and minima?
- What can you say about the first derivative when you have a local maximum? a local minimum?

Theorem (Local Extrema and Critical Points)

Suppose f is defined on an interval and has a local maximum or minimum at the point $x = a$, which is not an endpoint of the interval. If f is differentiable at $x = a$, then $f'(a) = 0$. Thus a is a critical point.

Theorem (First Derivative Test for Local Maxima and Minima)

Suppose p is a critical point of a continuous function f . Moving from left to right:

- If f' changes from negative to positive at p , then f has a local minimum at p .
- If f' changes from positive to negative at p , then f has a local maximum at p .

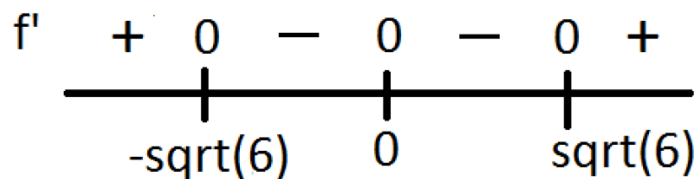
Find the critical points for $f(x) = x^5 - 10x^3 - 8$ and determine if each is a local maximum or minimum.

1. $f(x) = x^5 - 10x^3 - 8$

$$f'(x) = 5x^4 - 30x^2 = 5x^2(x^2 - 6) = 5x^2(x - \sqrt{6})(x + \sqrt{6}).$$

Therefore, the zeros of f' are $x = 0, \pm\sqrt{6}$ and these are the critical points.

Since f' is continuous, it cannot change sign between critical points. We can determine that on $(-\infty, -\sqrt{6})$, f' is positive either by picking a value (e.g. $f'(-3) = 135$) or by looking at the factored form (if $x < -\sqrt{6}$, $5x^2 > 0$, $(x - \sqrt{6}) < 0$ and $(x + \sqrt{6}) < 0$, so $(+)(-)(-) = (+)$).



At $x = -\sqrt{6}$, f' switches from positive to negative, so this is a local maximum. At $x = 0$, f' is negative on either side, so this is neither a local maximum nor a local minimum. At $\sqrt{6}$, f' switches from negative to positive, so this is a local minimum.